Feasibility Analysis of Airship-Deployed Hydrothermal Carbonization for Disaster Zone Waste Valorization

Technical Report
ME 401: Engineering Systems and Applications

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Disaster Relief - Waste Management - Thermal Energy - Airships

At a collection point outside of New Orleans, contractors process debris from Hurricane Katrina, Oct. 2005. *Office of History*

A flooded neighborhood in East Orleans Parish, Sept. 2005. *USACE photo by Hank Heusinkveld*

Project Idea

Idea:
Company that provides relief services following natural disasters.

↓

A modular system that can be delivered by airship

↓

HTC

Preliminary Question

Explore heat transfer for different sizes of reactors, fluid mechanics of process water, and contextual contstraints

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https://www.usace.army.mil/About/History/Historical-Vignettes/Relief-and-Recovery/170-Hurricane-Katrina/

Systems Design Strategy: Analysis Framework Decision

Energy Accounting

System must generate enough energy from waste conversion to power its own processing operations.

Energy Balance:
E_recovered >= E_reactor + E_preprocessing + E_auxiliary

Throughput vs. thermal mass — A small modular reactor loses proportionally more heat. Scale matters for closure.

Hydrochar quality vs. energy reinvestment — Burning your output product to power the reactor trades sequestration value for operational sustainability.

Moisture content vs. energy cost — HTC loves wet feedstock, but evaporating water is expensive. There's a crossover point where a feedstock becomes an energy sink rather than a source.

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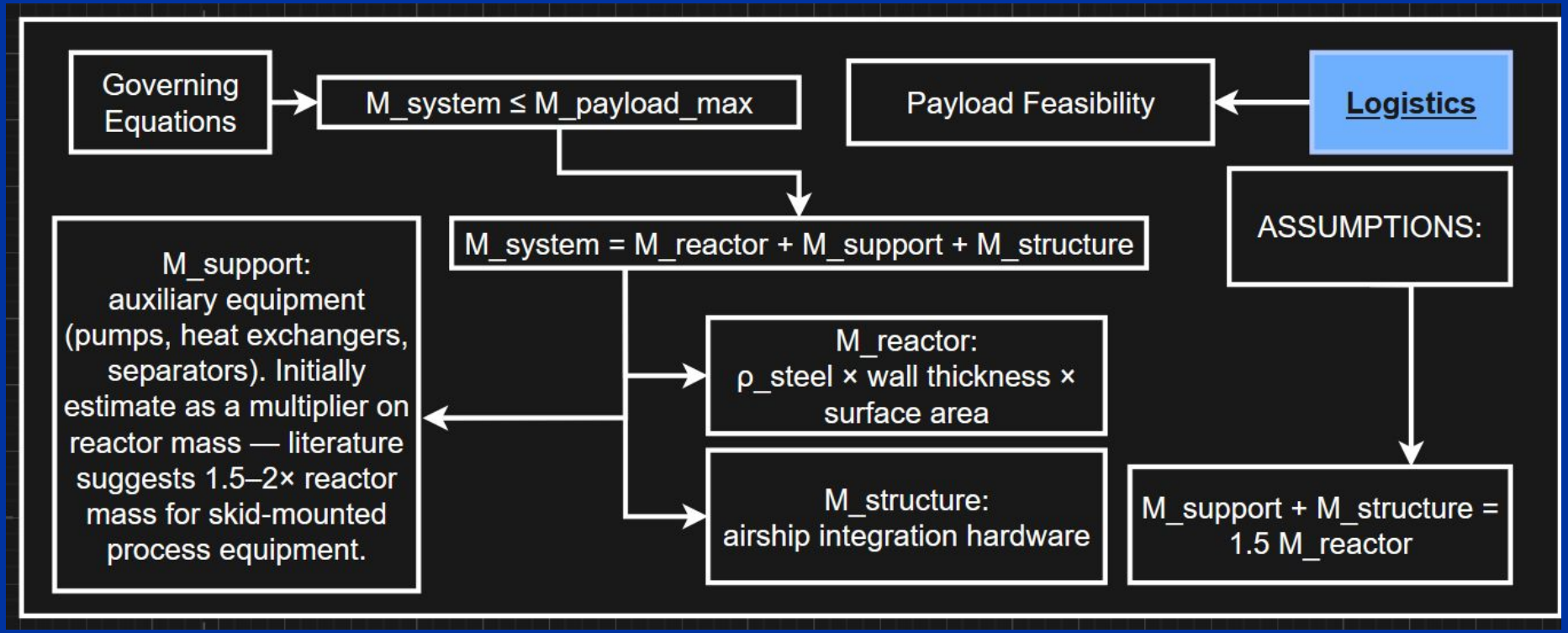
Analysis Map: Logistics



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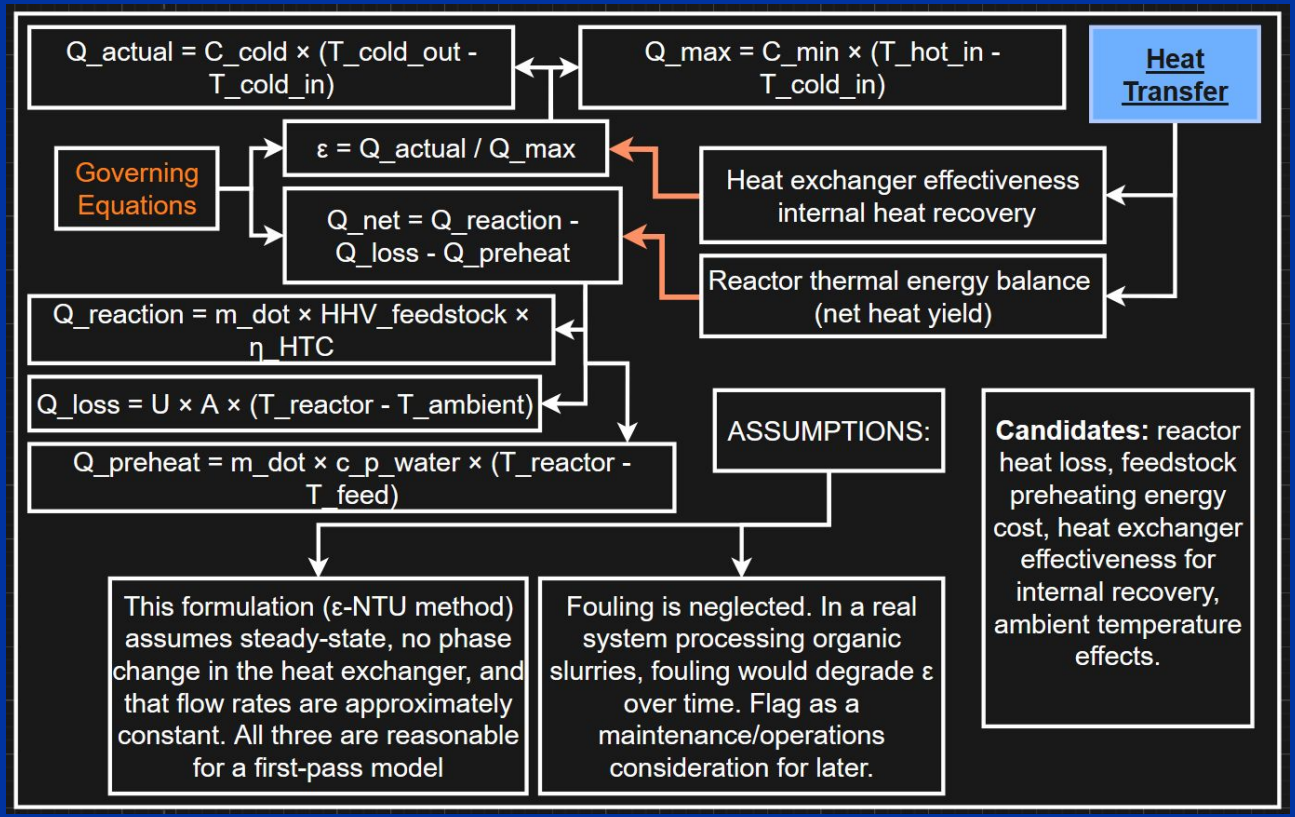
https://www.usace.army.mil/About/History/Historical-Vignettes/Relief-and-Recovery/170-Hurricane-Katrina/

Analysis Map: Heat Transfer



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https://www.usace.army.mil/About/History/Historical-Vignettes/Relief-and-Recovery/170-Hurricane-Katrina/

Analysis Map: Fluid Mechanics

ΔP_friction is estimated from the Darcy-Weisbach equation = $f \times (L/D) \times (\rho \times v^2/2)$

ΔP = ΔP_friction + ΔP_pressure_elevation (pump must overcome reactor pressure)

Governing Equations → W_pump = (m_dot_slurry × ΔP) / (ρ_slurry × η_pump)

Slurry Pump Parasitic Load ← **Fluid Mechanics**

ASSUMPTIONS:

Slurry rheology note: Organic slurries are typically non-Newtonian at high solids content (above ~15% solids by mass). The friction factor calculation assumes Newtonian behavior, which is valid for lower solids concentrations. At higher concentrations, the effective viscosity increases and the friction factor increases nonlinearly. Flag this: if your feedstock moisture analysis puts solids content above ~15%, the Newtonian assumption needs revisiting.

η_pump (pump efficiency) — use 0.6–0.7 for a first-pass estimate for centrifugal or progressive cavity pumps handling slurries.

Autogenous pressure is a function of temperature — you need the saturation pressure of water at your chosen HTC operating temperature. This is directly available from steam tables.

Pipe geometry (L, D) is a design parameter. Initial estimate: assume compact skid layout, total pipe run ~10–15 meters, diameter sized for ~1–3 m/s slurry velocity.

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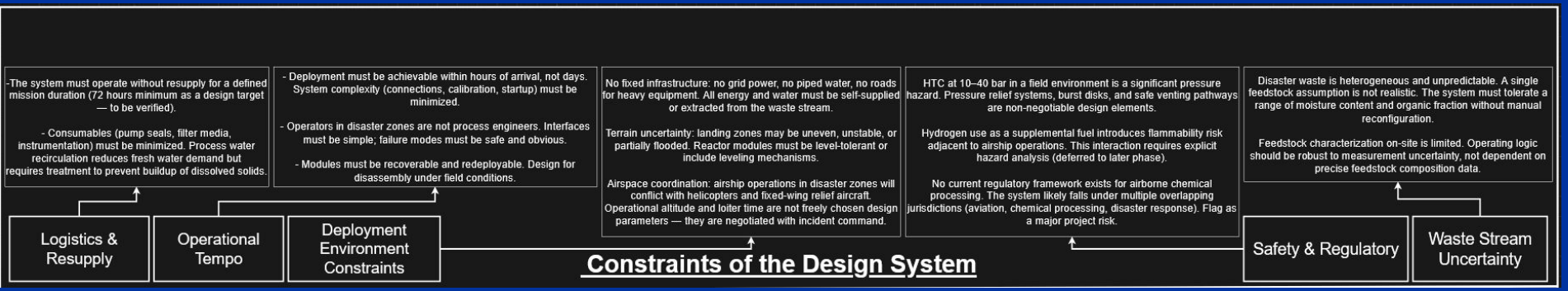
System Map: Constraints



Key Constraints:

- <u>All energy and water must be self-supplied or extracted from the waste stream.</u>
- Disaster waste is heterogeneous and unpredictable. A single feedstock assumption is not realistic. The system must tolerate a range of moisture content and organic fraction without manual reconfiguration.
- HTC at 10–40 bar in a field environment is a significant pressure hazard. Pressure relief systems, burst disks, and safe venting pathways are non-negotiable design elements.

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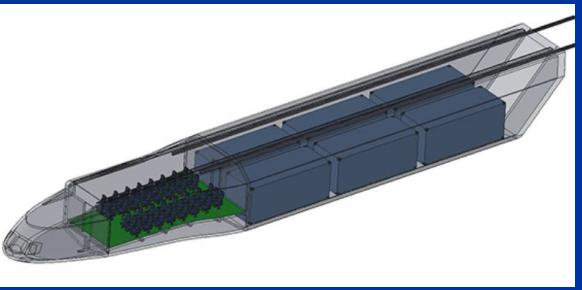
System Map: Design Modules



Module 1
Payload

Primary Equations:

Buoyant force: $F_b = (\rho_{air} - \rho_{lift_gas}) \times V_{envelope} \times g$

Net lift: $F_{net} = F_b - M_{airship_empty} \times g$

Payload budget: $M_{payload} = F_{net} / g$

Module count: $N_{modules} = \text{floor}(M_{payload} / M_{per_module})$

Hydrogen Offtake:

ΔM_{H2_vented} = mass of hydrogen vented after deployment

$\Delta F_{b_lost} = \Delta M_{H2_vented} \times (\rho_{air} - \rho_{H2}) \times g$

$E_{H2_gained} = \Delta M_{H2_vented} \times LHV_{H2}$

Inputs:
Envelope volume (m^3), lift gas identity (H2 or He), airship empty mass (kg), module mass estimate (kg/module)

Outputs:
Maximum module count N; feasible throughput range (kg/hr); hydrogen offtake energy available vs. buoyancy cost curve

Module 2
Reactor Thermal Balance

Primary Equations:

$Q_{net} = Q_{reaction} - Q_{loss} - Q_{preheat_net}$

$Q_{reaction} = m_{dot} \times HHV_{feedstock} \times \eta_{HTC}$

$Q_{loss} = U \times A_{reactor} \times (T_{reactor} - T_{ambient})$

$Q_{preheat} = m_{dot} \times moisture_{fraction} \times c_{p_water} \times (T_{reactor} - T_{feed})$

$Q_{preheat_net} = Q_{preheat} \times (1 - \varepsilon)$ [after heat exchanger recovery]

Inputs:
m_{dot} (from Module 1), HHV and η_{HTC} (from literature per feedstock), $moisture_{fraction}$ (swept 0.40–0.80), $T_{reactor}$ (design parameter, 180–250°C), U (estimated $W/m^2 \cdot K$), $A_{reactor}$ (from Module 3 geometry), ε (from Module 4), $T_{ambient}$, T_{feed}

Outputs:
Q_{net} (kW) as function of moisture and throughput; crossover moisture threshold ($Q_{net} = 0$ boundary); minimum η_{HTC} for self-sufficiency at each moisture level

Key parameters to obtain from literature before coding:
HHV (kJ/kg dry) and η_{HTC} for municipal food waste and sewage sludge

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https://www.hybridairvehicles.com/

System Map: Design Modules

Module 3
Pressure Vessel

Primary equations:

Hoop stress: $\sigma_{hoop} = P \times r / t$

Required thickness: $t = P \times r / (\sigma_{allowable} \times \eta_{weld})$

Vessel mass: $M_{vessel} = \rho_{steel} \times 2\pi \times r \times L \times t$ (cylindrical shell)

End caps add approximately 15–20% to shell mass (first-pass estimate)

Where:

P = autogenous pressure at T_reactor (from steam tables, Pa)
r = inner radius of vessel (m), a design parameter
$\sigma_{allowable}$ = yield strength / safety factor (Pa); for 316 SS, $\sigma_{yield} \approx 205$ MPa; apply safety factor of 3–4 for pressure vessel service → $\sigma_{allowable} \approx 50$–$68$ MPa
η_{weld} = weld efficiency factor, typically 0.85 for radiographed welds
$\rho_{steel} \approx 8000$ kg/m³

Inputs:
T_reactor (sets P via steam tables), vessel geometry (r, L), material selection (σ_{yield}, ρ), safety factor, weld efficiency

Outputs:
Wall thickness t (mm); vessel mass M_{vessel} (kg); M_{vessel} as function of T_reactor; how vessel mass changes module count (feeds back into Module 1)

Key tension:
Higher T_reactor → higher P → thicker walls → heavier vessel → fewer modules per airship. This coupling must be explicit.
Design decision produced: Operating temperature that balances reaction kinetics benefit against structural mass penalty

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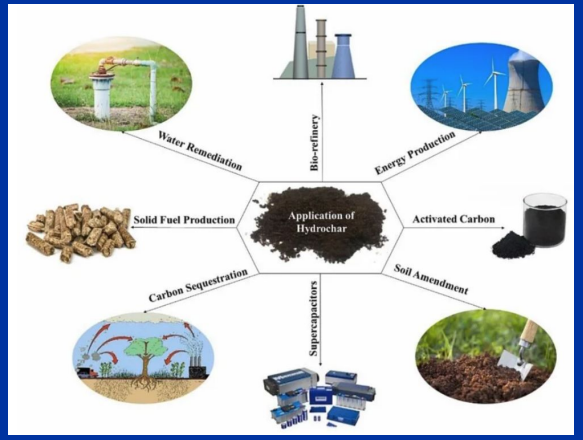
https://rsellc.com/products/pressure-vessel/

System Map: Design Modules



Module 4
Heat Exchanger Efficiency

Primary equations (ε-NTU method):

$\varepsilon = Q_{actual} / Q_{max}$

$Q_{max} = C_{min} \times (T_{hot_in} - T_{cold_in})$

$C = m_{dot} \times c_p$ for each stream

$C_{min} = \min(C_{hot}, C_{cold})$

$Q_{recovered} = \varepsilon \times C_{min} \times (T_{hot_in} - T_{cold_in})$

Inputs:
ε (design parameter, swept 0.0–0.85), T_hot_in (reactor outlet, ~T_reactor), T_cold_in (feed temperature, ~15–30°C), m_dot, c_p for each stream (~4.18 kJ/kg·K for water-dominant streams)

Outputs:
Q_recovered (kW); minimum ε to close energy balance; sensitivity of Q_net to ε degradation (fouling scenario); Q_net at ε=0 (no-recovery baseline) vs. ε=0.70 (realistic)

Assumption:
Steady-state counterflow, no phase change in HX, constant flow rates. Fouling neglected at this analysis stage.

Module 5
Self-Sufficiency Integration

Governing inequality (assembled from Modules 2, 3, 4):

$Q_{net}(moisture, m_{dot}, T_{reactor}, \varepsilon) >= W_{pump} + W_{auxiliary}$

Pump work (from slurry mechanics):

$W_{pump} = m_{dot} \times \Delta P / (\rho_{slurry} \times \eta_{pump})$

ΔP dominated by $P_{autogenous}(T_{reactor})$ — pressurization cost

Darcy-Weisbach for friction: $\Delta P_f = f \times (L/D) \times (\rho \times v^2 / 2)$

$\eta_{pump} = 0.65$ (initial estimate for slurry service)

$\rho_{slurry} \approx 950–1100$ kg/m³ (function of moisture and solids density)

W_auxiliary: estimated at 5–10% of Q_net as placeholder for instrumentation, controls, communications load

Output:
2D feasibility surface in {moisture content × throughput} space showing:
(1) the self-sufficient operating region, (2) the boundary Q_net = W_pump + W_auxiliary, (3) sensitivity to ε and T_reactor as parameter shifts.
This surface is the primary technical deliverable of the project.

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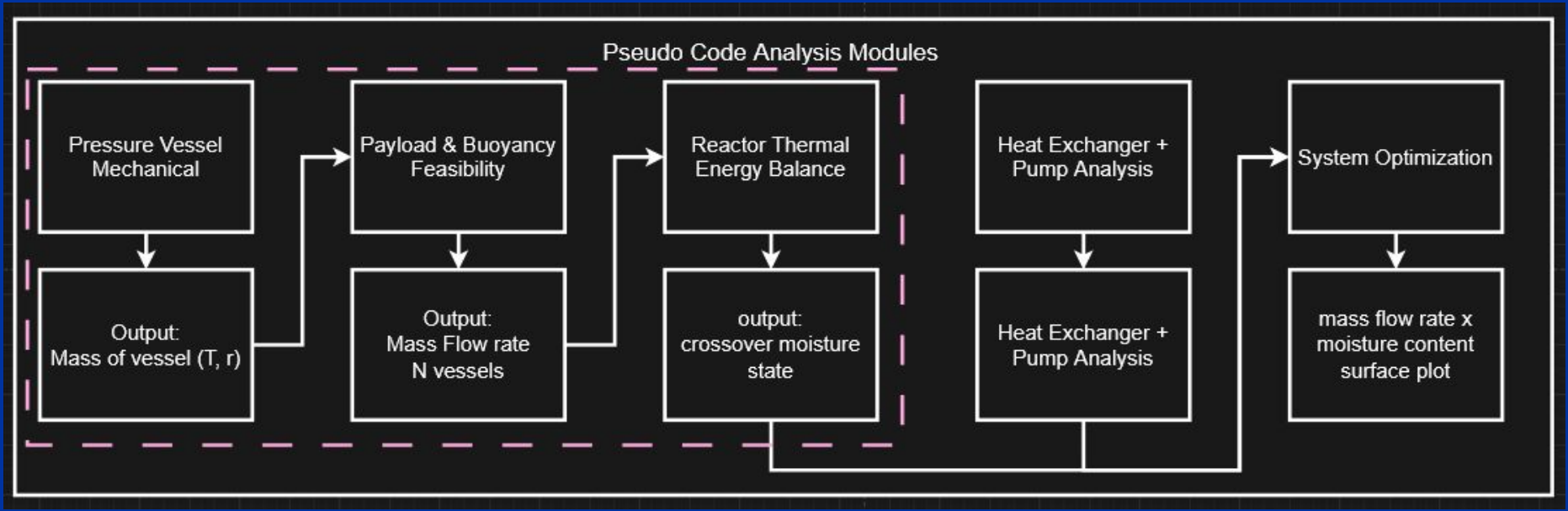
https://www.researchgate.net/figure/Overview-of-different-applications-of-hydrochar_fig1_391394878

Pseudo Code



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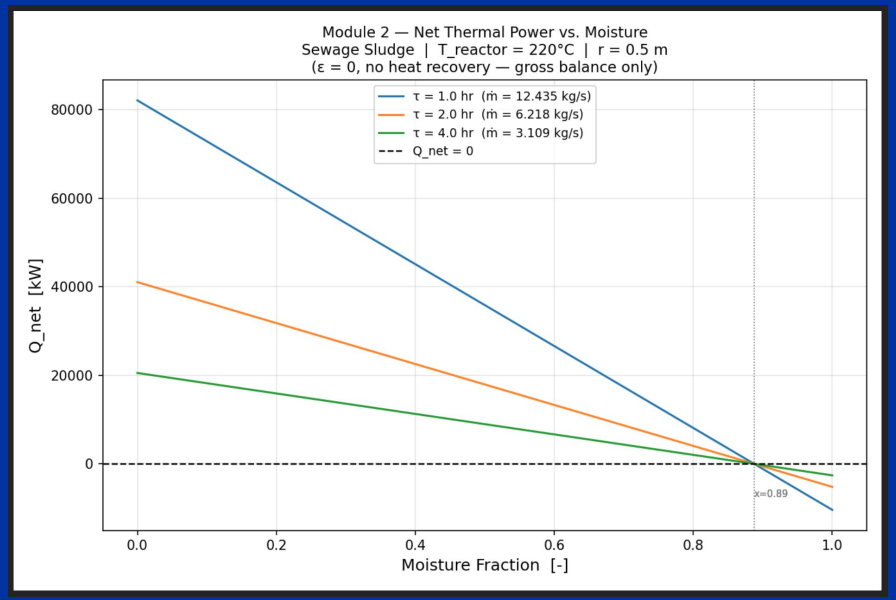
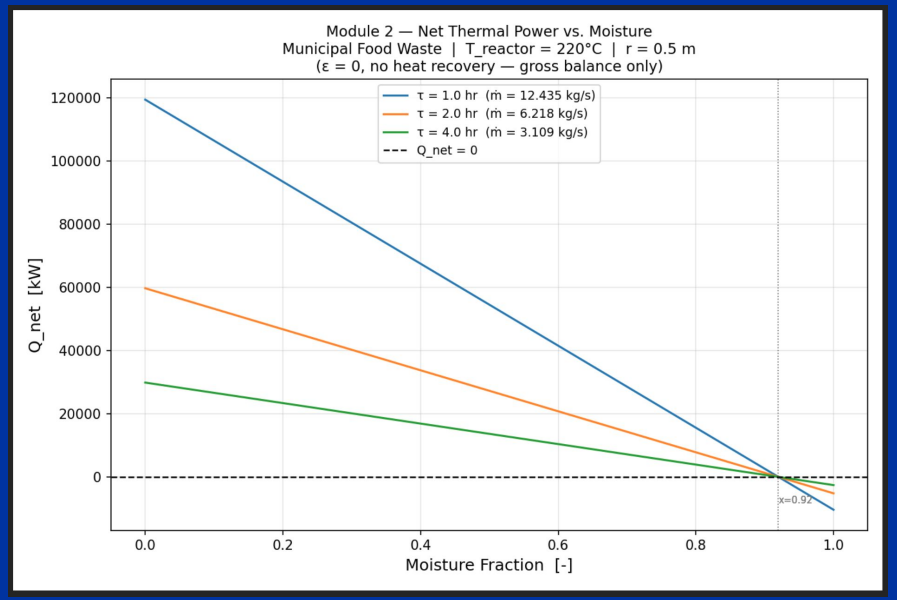
Results

First Pass Analysis favor mass flow rate. Thermal Module Development is expected to change this





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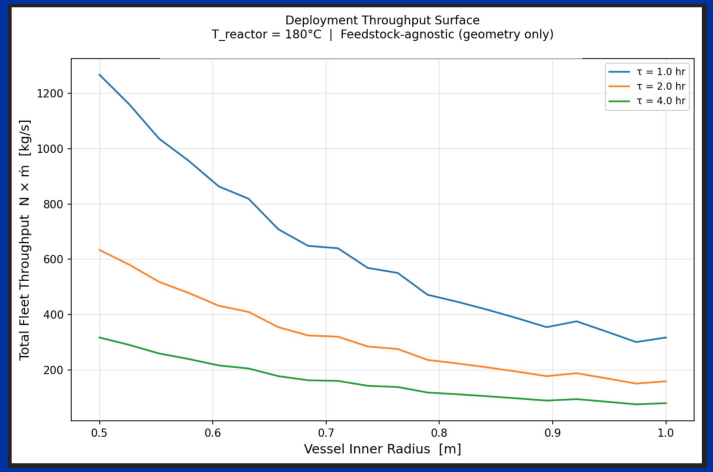
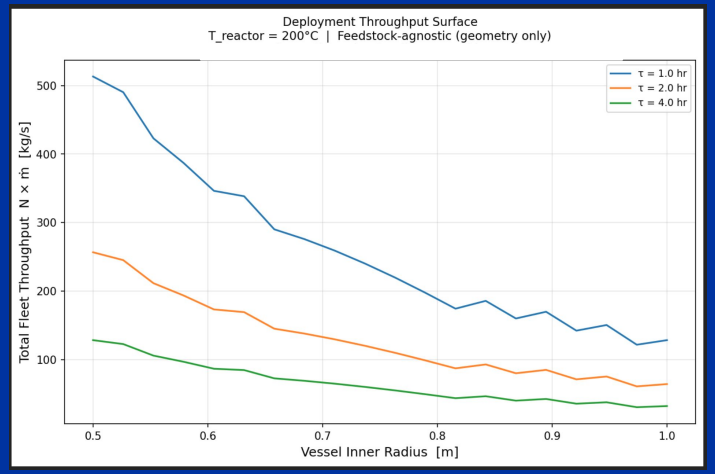
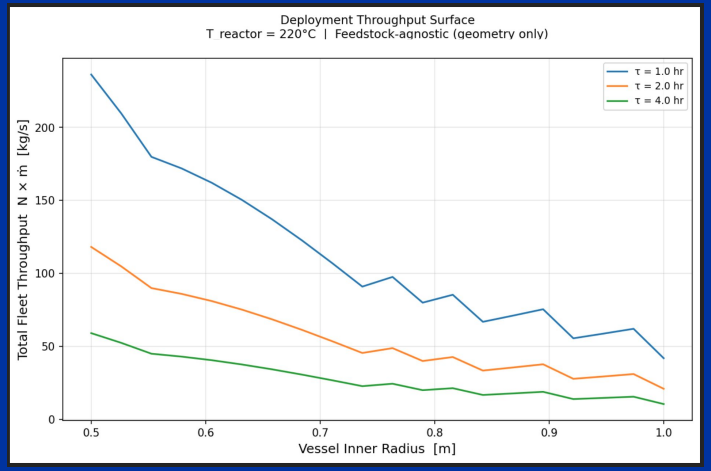
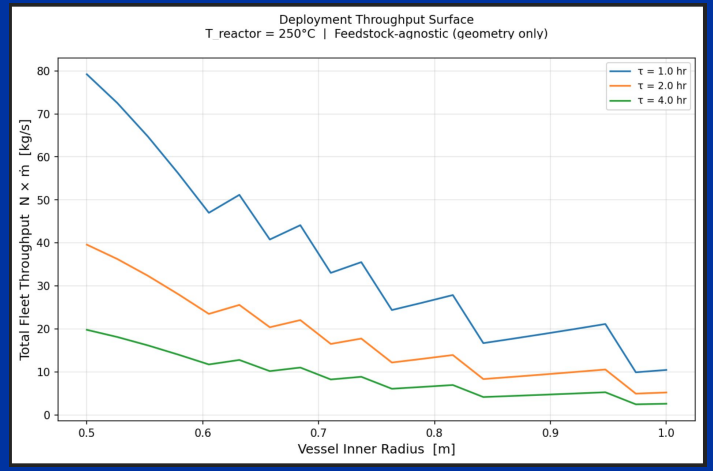
Results

Resource Deployment is a function of required reaction temperature, feedstock moisture, and throughput desire









Deployment packages could be tailored to optimize feedstock type, throughput, intended products.

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